

Riley Beenders

ENGINEERING LEADERSHIP | SOFTWARE-INTEGRATED SYSTEMS | TECHNOLOGY STRATEGY

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PROFESSIONAL SUMMARY

Engineering and technology leader with 10+ years of multidisciplinary experience, including CTO and Lead R&D roles spanning hiring, vendor partnerships, and technology strategy. Delivers software-integrated products and internal business systems, including a self-developed ERP/CRM on a redundant Kubernetes cluster, remote operations and analytics platforms, and Python, LabVIEW, and TypeScript/Node.js applications, from requirements through reliable operation and support.

CORE EXPERTISE

Leadership & Delivery: Technical team leadership; hiring and technical interviews; external vendor and software-partner management; vendor cost negotiation; technology strategy; requirements definition; worldwide collaboration

Software: Python; TypeScript/React/Next.js (Node.js runtime); LabVIEW; Git/GitHub; ERP/CRM application development; Modbus data integration; application testing and debugging

Platforms & Reliability: Kubernetes; Docker; Proxmox virtualization; Windows/Linux administration; Vercel deployment; redundant self-hosted services; metrics, live monitoring, and web analytics; risk analysis

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions while leading R&D product development from requirements through commercialization.
- Implemented ProteorPrint OS with external software partner SimplyPrint, enabling remote operation, AI-assisted diagnostics, live video, and material, waste, and outcome metrics.
- Managed the Coplay site's Microsoft Business Central ERP implementation, syncing SKUs with Proteor's global locations, and standardized software and internal processes after the acquisition.
- Built an automated LabVIEW QC application that negotiates Modbus sensor communication, charts live tolerances, and generates traceable PDF reports linked to QR-coded product IDs.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Progressed through four roles to CTO, overseeing the product-development pipeline, internal IT infrastructure, and technology strategy.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized orders, inventory, tasks, projects, and customer interactions and automated dynamic BOM generation linked to purchasing.
- Coordinated vendors on custom harnesses, panel designs, and hardware improvements, and led large automated systems from CAD and BOMs through firmware tuning, customer support, and operator training.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging for larger, modular machines.

ADDITIONAL RELEVANT EXPERIENCE

RileyBeenders.com - Full-Stack Portfolio Project | 2026 | Built and deployed a TypeScript/React/Next.js site on Vercel with structured JSON content, a Node.js PDF endpoint, Git version control, a GitHub Actions workflow, and web analytics.

RB's Auto App Updater - PowerShell Automation Project | 2026 | One-line PowerShell install that upgrades every WinGet package via an elevated scheduled task, maps exit codes to plain-language statuses, and self-updates after SHA-256 verification.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks Machine Learning and MATLAB Onramps (2026); Lean Six Sigma Yellow Belt (2025)